2 Goal and Area of Application

Phospholipidosis is a storage disorder of excessive accumulation of phospholipids in lysosomes that can be drug-induced. understanding the role of drug-induced phospholipidosis (DIPL) in ongoing drug screenings is necessary to better to distinguish between on-target and side effects as well as for putative endosomal pathway disturbance.

3 Definitions and Abbreviations

SOP	**S**tandard **O**perating **P**rocedure
SP	**S**creening **P**ort
ITMP	**I**nstitut für **T**ranslationale **M**edizin und **P**harmakologie
RT	**R**oom **T**emperature (20-25 °C)
i.a.	**I**n **A**ssay
DIPL	**D**rug **I**nduced **P**hospho**l**ipidosis
DMEM	**D**ulbecco's **M**odified **E**agle **M**edium
HCS	**H**igh **C**ontent **S**creening
PBS	**P**hosphate-**B**uffered **S**aline
FCS	**F**etal **C**alf **S**erum
DMSO	**D**i**m**ethyl **s**ulf**o**xide
EDTA	**E**thylene**d**iamine**t**etraacetic **a**cid
ms	**m**illi**s**econds

4 Method

4.1 General

This SOP describes the procedure of measuring Drug-Induced-Phospholipidosis staining in VeroE6 cell line in combination with an Operetta HCS system (Revvity) at Fraunhofer ITMP-SP. HCS LipidTOX™ Red Phospholipidosis Detection Reagent (invitrogenTM, #H34351) is a fluorescent dye that is used to stain the cells and to detect induction of phospholipidosis. The dye does not influencing cell growth and is well-retained after aldehyde fixation.

4.2 Information for consumables / chemicals / equipment

Product Name	Supplier	Catalog Number
HCS LipidTOX™ Red Phospholipidosis Detection Reagent (1000X), for cellular imaging	Invitrogen TM	H34351
CellMask™ plasma membrane stain	Invitrogen TM	C10046
Hoechst 33258	Sigma Aldrich Inc., MO, USA	94403
e.g. Formaldehyde solution (PFA)	Sigma Aldrich Inc., MO, USA	252549
Amiodarone	Sigma Aldrich Inc., MO, USA	V0627

DMEM; high glucose, no L-glutamine, no phenol red	Life Technologies GmbH	31053044
Vero E6	ATCC, USA	CRL-1586
e.g. PBS w/o Ca^{2+}, Mg^{2+}	Gibco, Life Technologies GmbH	14190250
e.g. Trypsin 0.05 % / EDTA 0.02% in PBS w/o Ca^{2+}	Capricorn	TRY-1B
e.g. FCS	Capricorn	10-FBS-11F
e.g. Penicillin 10,000 U/mL / Streptomycin 10,000 µg/mL	Gibco, Life Technologies GmbH	15140122
e.g. Glutamine	Gibco, Life Technologies GmbH	25030081
PhenoPlate 384-well, black, optically clear flat-bottom, tissue-culture treated	Revvity	6057302
DMSO	Carl Roth, Germany	HN47.1
Echo	Labcyte Inc., CA, USA	
Operetta CLS High-Content Analysis System	Revvity	HH16000020

4.3 Procedure

1. Compound addition

   Compounds (10 µM i.a. concentration) and controls (amiodarone 10 µM i.a. concentration; DMSO 0.1 v/v %) are added using the Echo Liquid Handler (Labcyte). Due to requests of the Echo system, attention should be paid to the selected plate types to avoid the plate and transducer crash.

   Important: PhenoPlate = CellCarrier plate in the Echo software

2. Cell seeding

   VeroE6 cell line is used for detection of DIPL. Cells are culture in complete medium: DMEM (Dulbecco's Modified Eagle Medium) supplemented with 10% FBS, 6 mM L-glutamine, 1% penicillin/streptomycin. Cells were harvested from a T175 cm^2 flask at 80% confluence by washing them once using 10 mL RT PBS and afterwards incubating the cells with 5 mL Trypsin 0.05% / EDTA 0.02 % for 5 min. Cells are suspended using 10 mL of prewarmed cell culture media. VeroE6 cells are diluted to a concentration of 500,000 cells per mL in cell culture media supplemented with 1:1000 diluted HCS LipidTOX™ Red Phospholipidosis Detection Reagent (1000X). 20 µL of this suspension is added to each well of a 384 well PhenoPlate. Cells are incubated for 24 h at 37 °C and 5 % CO_2.

3. Detection

Cells are fixed by addition of 20 µL/well 8 % formaldehyde solution. After 30 min incubation time at RT fixation buffer is removed and cells are washed with 50 µL/well PBS. Cells are stained for 45 min with 20 µL/well of CellMask™ plasma membrane stain (1:3,000) and Hoechst 33258 (1:10,000) diluted in PBS. Staining solution was removed and cells were washed 3 times with 50 µL/well PBS.

Important: Increased fixation-time may result in loss of signal (DIPL spots) and enhanced background.

Induction of DIPL was detected at 0 µm height of detection with 5 images/well on the Operetta CLS High-Content Analysis System using following parameters.

Channel-name	Power	Exposure time
Alexa 594	60 %	20 ms
Hoechst 33258	100 %	20 ms
CellMask Deep red	100 %	80 ms

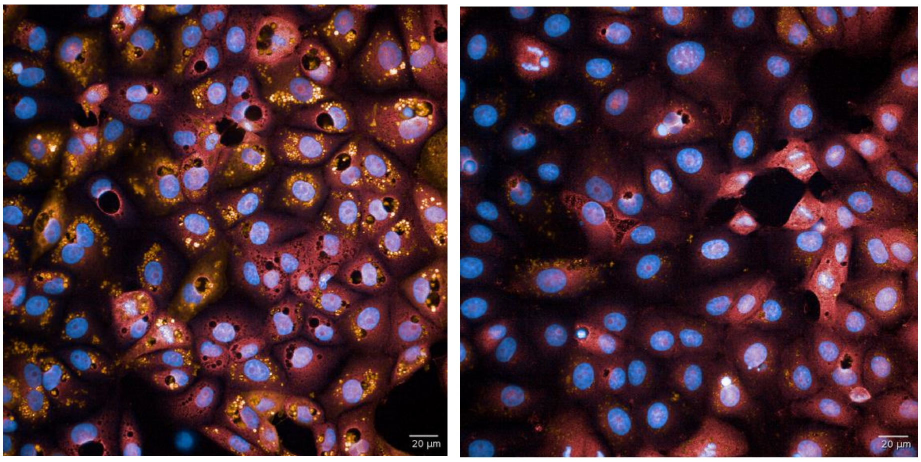
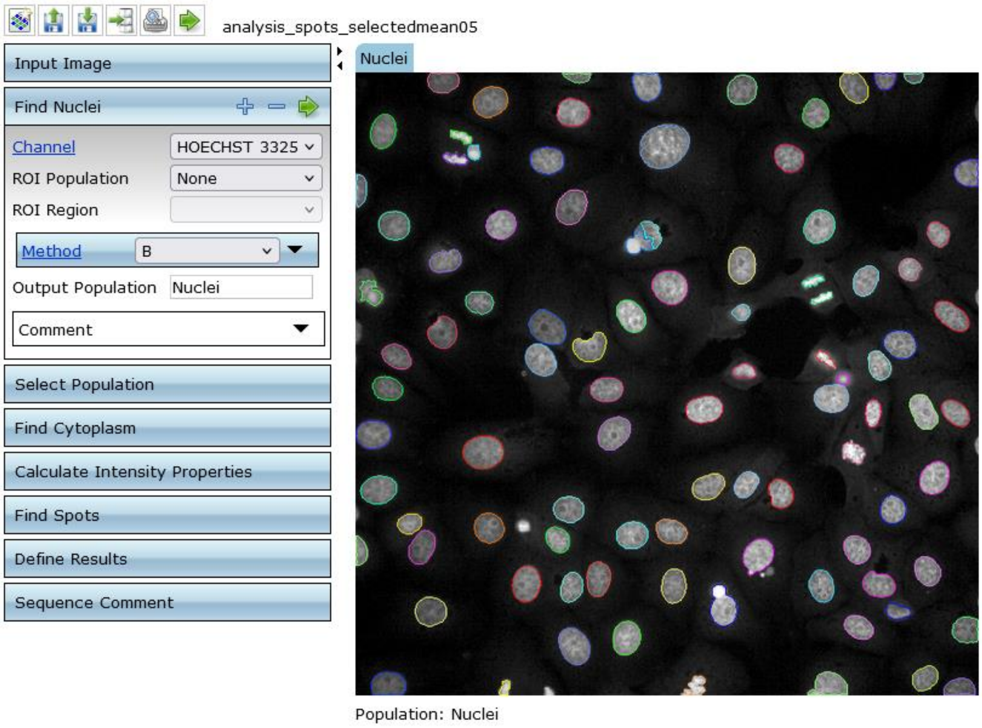


Fig. 1. Example of a positive control (left, amiodarone 20 µM) and negative control (right, DMSO 0.1 v/v %) effect. Nuclei stain using Hoechst 33258, cytoplasm stain using CellMask Deep Red.

4. Analysis

Image analysis is performed using Columbus image analysis software. The analysis includes 5 steps:

- Nuclei identification (Fig.1-A)
- Removement of border objects (nuclei are completely displayed) = “selected population” (Fig.1-B)
- Cytoplasm identification in the selected population (Fig.1-C)
- Calculation of DIPL signal in the cytoplasm (intensity) (Fig.1-D)
- Identification of spots (Fig.1-E)
- Definition of results that should be calculated (Fig.1-F)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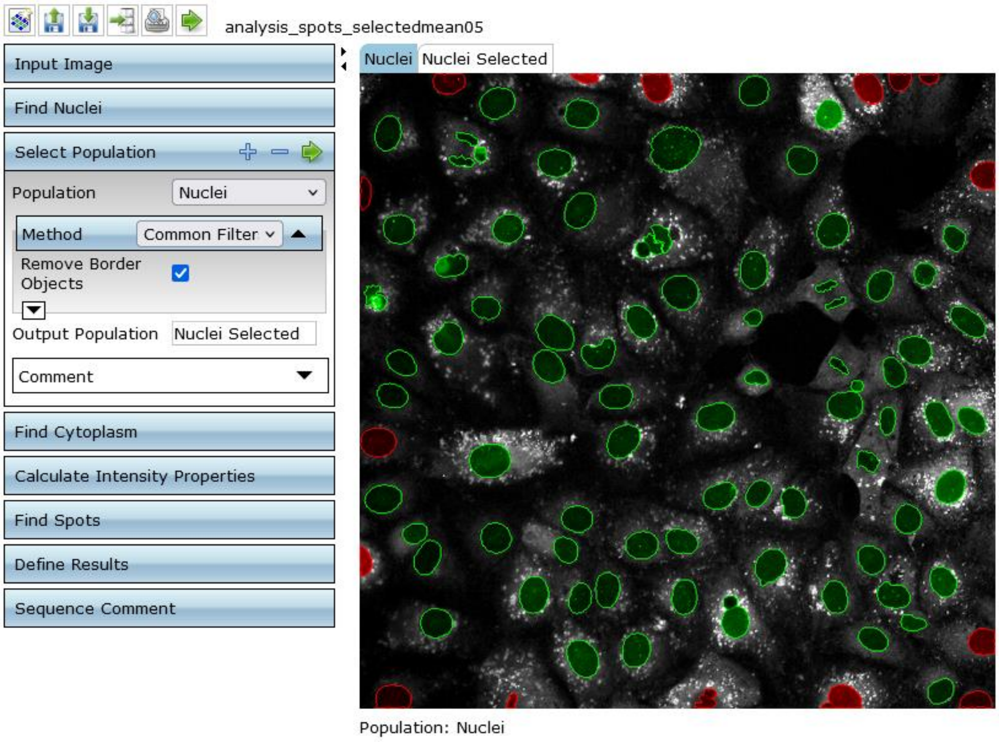
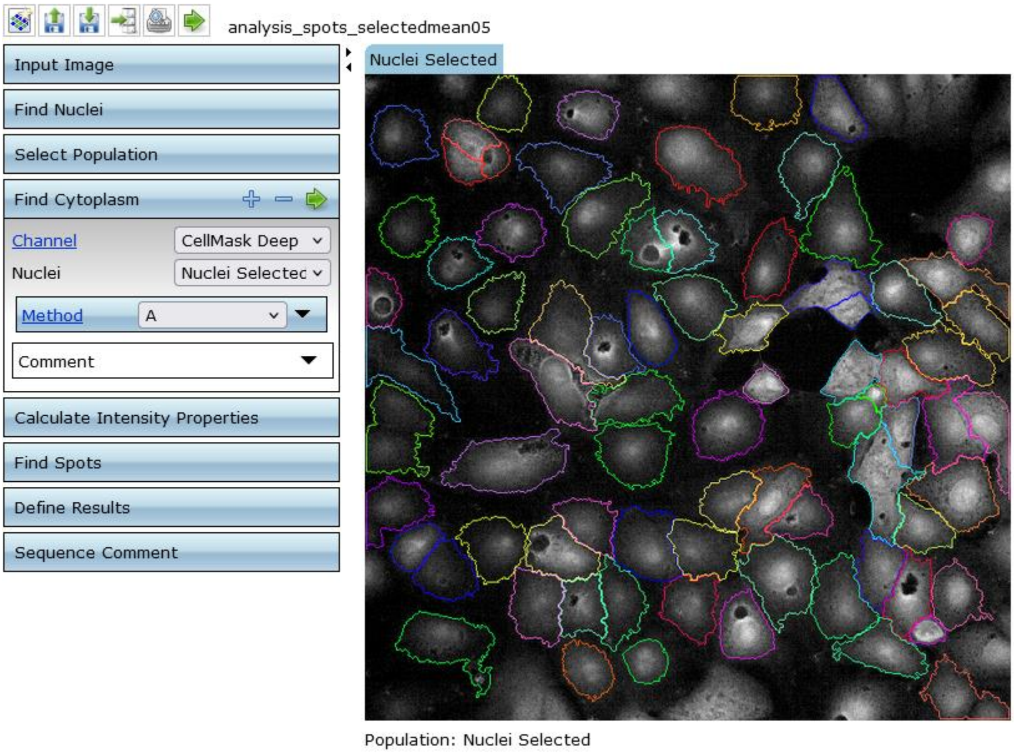
Input Image
Find Nuclei
Channel
HOECHST 3325
ROI Population
None
ROI Region
Method
B
Output Population
Nuclei
Comment
Select Population
Find Cytoplasm
Calculate Intensity Properties
Find Spots
Define Results
Sequence Comment
Nuclei
Population: Nuclei

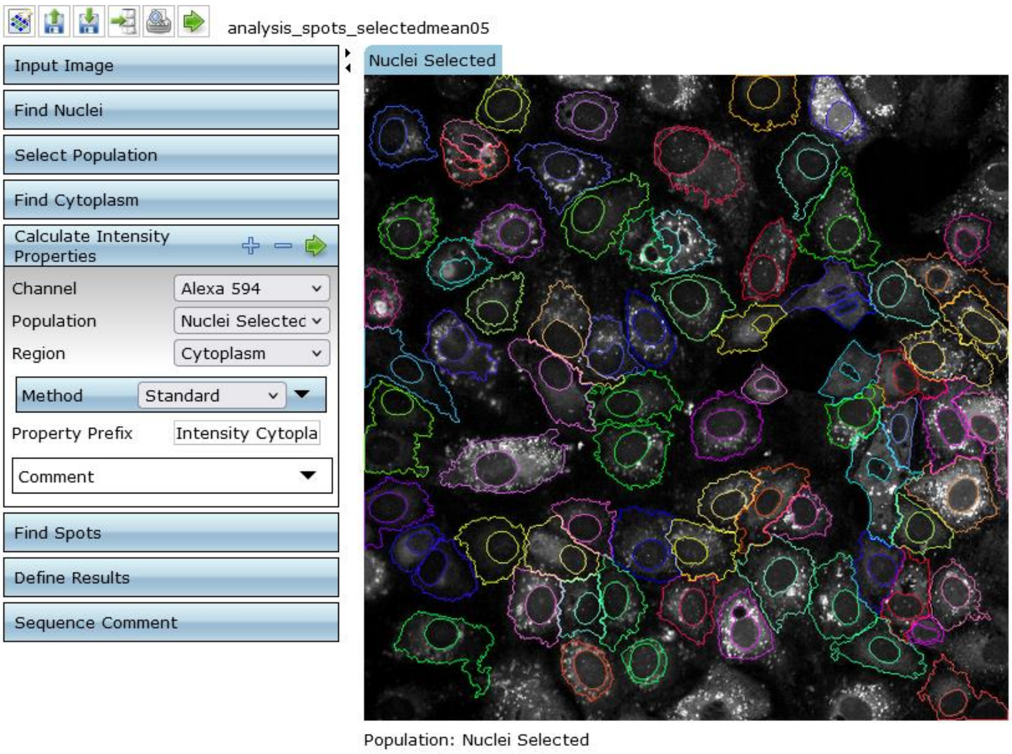
A


analysis_spots_selectedmean05
Input Image
Find Nuclei
Select Population
Population
Nuclei
Method
Common Filter
Remove Border Objects
Output Population
Nuclei Selected
Comment
Find Cytoplasm
Calculate Intensity Properties
Find Spots
Define Results
Sequence Comment
Nuclei
Nuclei Selected
Population: Nuclei

B

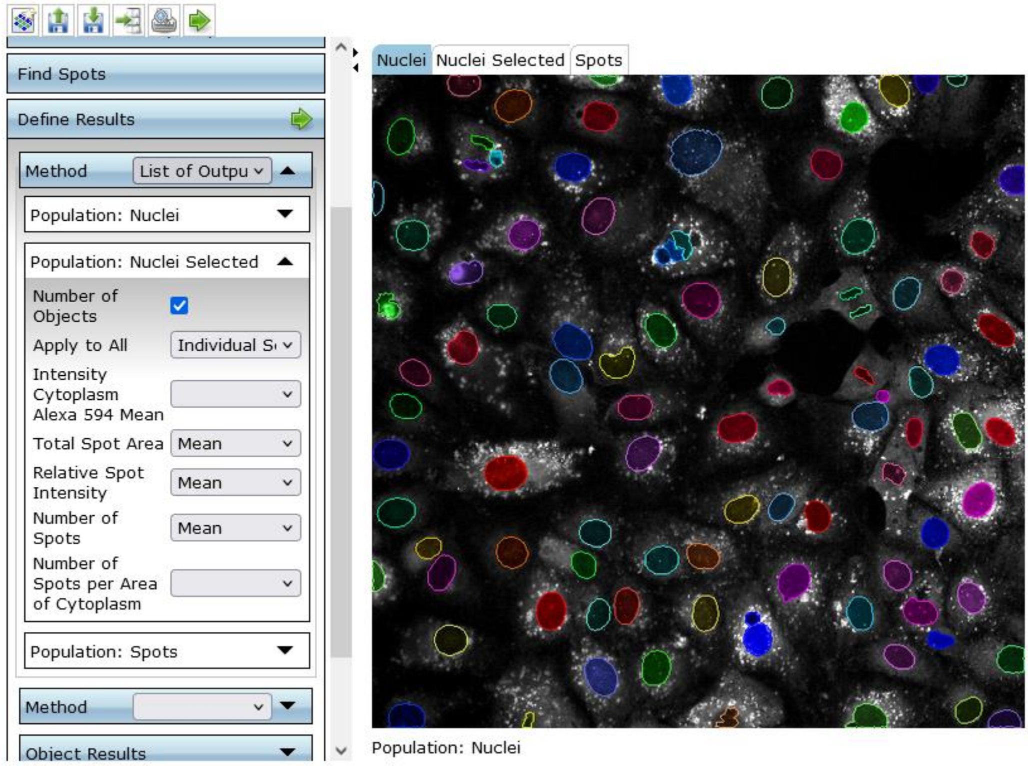

analysis_spots_selectedmean05
Input Image
Find Nuclei
Select Population
Find Cytoplasm
Channel
CellMask Deep
Nuclei
Nuclei Selectec
Method
A
Comment
Calculate Intensity Properties
Find Spots
Define Results
Sequence Comment
Nuclei Selected
Population: Nuclei Selected

C


analysis_spots_selectedmean05
Input Image
Find Nuclei
Select Population
Find Cytoplasm
Calculate Intensity Properties
Channel
Alexa 594
Population
Nuclei Selectec
Region
Cytoplasm
Method
Standard
Property Prefix
Intensity Cytopla
Comment
Find Spots
Define Results
Sequence Comment
Nuclei Selected
Population: Nuclei Selected

D

E



F

Figure 1. Workflow of DIPL data analysis in Columbus Image Analysis Software. Nuclei identification (A) Removement of border objects (nuclei are completely displayed) = "selected

population" (B) Cytoplasm identification in the selected population (C) Calculation of DIPL signal in the cytoplasm (intensity) (D) Identification of spots (E) Definition of results that should be calculated (F)

5 Further applicable resources

Manufacturer manual for HCS LipidTOX™ Red Phospholipidosis Detection Reagent (1000X), for cellular imaging:

- https://assets.thermofisher.com/TFS-Assets%2FLSG%2Fmanuals%2Fmp34350.pdf

6 Document history

Guidance: Record the version history and reasons for updates.

SOP version	**Reason**
Version 1	Initial draft of document
Version 2	*E.g., update of section 7*
Version 3	*E.g., correction of typos*

7 Annex

No annex